GEOMETRY · CIRCLES

Circle Angle Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Circle Angle Explorer** on Mr. K's Classroom: Geometry → Circles → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. A central angle and an inscribed angle intercept the same 100° arc. Which is bigger, and by how much?

   The central angle: 100° versus 50°.

2 Explore use the tool and record what you see

Central and inscribed angles

Drag A and B to set each arc.

Arc AB	Central ∠AOB	Inscribed ∠APB
120°	**120°**	**60°**
100°	**100°**	**50°**
60°	**60°**	**30°**
180° (AB is a diameter)	**180° (a straight line)**	**90°**

Arc length and sector area

Use the second tab. Leave answers in terms of π.

r	Angle	Fraction of circle	Arc length	Sector area
4	90°	$\frac{1}{4}$	**2π**	**4π**
6	60°	$\frac{1}{6}$	**2π**	**6π**
3	120°	$\frac{1}{3}$	**2π**	**3π**
5	180°	$\frac{1}{2}$	**5π**	$\frac{25}{2}\pi$

3 Write the rule fill in each blank from what you observed

1. An inscribed angle is **half** the measure of its intercepted arc.
2. A central angle is **equal to** its intercepted arc.
3. An inscribed angle that intercepts a semicircle measures **90°**.
4. Inscribed angles that intercept the same arc are **congruent**.
5. Arc length = $\frac{\text{angle}}{360°}$ · **2πr** and sector area = $\frac{\text{angle}}{360°}$ · **πr²**.

4 Check no tool: show your work

1. An inscribed angle measures 35°. Find its intercepted arc.

   70°

2. Find the arc length when r = 9 and the central angle is 40°.

   2π

3. Find the area of a sector with r = 10 and a 72° central angle.

   20π

4. Triangle ABC is inscribed with AB a diameter and ∠A = 25°. Find ∠B.

   65° (∠C = 90°)

5 Reflect

Why doesn't the inscribed angle change when P slides along the circle?

It always intercepts the same arc, and the angle is always half that arc.